

# Software Requirements Specification for LeadOS

## 1. Introduction

### 1.1 Product Name

LeadOS

### 1.2 Product Vision

LeadOS is a multilingual AI lead conversion platform that helps businesses convert raw campaign leads into qualified sales opportunities using AI chat, AI voice, structured lead scoring, RM handoff, WhatsApp follow up links, and dashboard analytics.

The MVP will focus on a fictional financial partner program called FinPartner Pro so that the product can be demonstrated safely without using Rupeezy branding or protected IP.

### 1.3 Purpose of this Document

This SRS defines the complete software requirements for the LeadOS MVP.

It covers:

- 1. Product scope
- 2. User roles
- 3. Portals and websites
- 4. Functional requirements
- 5. Data requirements
- 6. AI requirements
- 7. Scoring requirements
- 8. User flows
- 9. System flows
- 10. Non functional requirements
- 11. Security requirements
- 12. Integration requirements
- 13. MVP boundaries
- 14. Acceptance criteria

This document should guide the team during development, testing, and demo preparation.

### 1.4 Product Goal

The goal of LeadOS MVP is to prove the following flow:

```
Lead submits form
→ AI starts chat or voice conversation
→ AI detects language
→ AI explains partner program
→ AI handles objections
→ AI qualifies the lead
→ Transcript is saved
→ Scoring runs automatically
→ Lead becomes Hot, Warm, or Cold
→ Hot lead goes to RM with full context
→ Warm lead gets WhatsApp follow up link
→ Cold lead is closed
→ Admin dashboard updates
```

### 1.5 MVP Focus

The MVP must focus only on the theme requirements and our submitted proposal.

The system should not become a full CRM, HRMS, marketing automation tool, or enterprise workflow engine. It should prove AI driven lead conversion clearly and reliably.

## 2. Overall Product Description

### 2.1 Product Summary

LeadOS is an enterprise SaaS platform with four main user facing surfaces:

1. LeadOS SaaS Website
2. Dummy Client Website
3. Admin Portal
4. RM Portal

The backend will connect these surfaces through Supabase Edge Functions, Supabase Auth, Supabase Postgres, AI conversation services, scoring logic, routing logic, and follow up workflows.

## 2.2 Main Problem

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Businesses often lose leads because:

1. Leads are not contacted quickly
2. RMs can handle only one conversation at a time
3. Campaign spikes create lead backlog
4. Language mismatch causes early drop off
5. RMs do not get enough context from previous conversations
6. Objections are handled inconsistently
7. Follow up is not structured
8. Managers lack visibility into funnel quality

LeadOS solves this by placing an AI lead conversion layer before the human RM.

## 2.3 Product Outcome

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LeadOS should help the business:

1. Contact leads faster
2. Speak in the lead's preferred language
3. Handle common objections consistently
4. Score every lead explainably
5. Route Hot leads to RMs
6. Send Warm leads follow up links
7. Reduce repetitive top funnel work
8. Give admins visibility into lead quality and conversion funnel

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## 3. User Roles

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### 3.1 Lead

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The Lead is the external person who fills the form on the dummy client website and interacts with the AI assistant.

The Lead can:

1. View dummy client partner program page
2. Fill lead form
3. Select preferred language
4. Select preferred channel
5. Give WhatsApp consent
6. Start chat
7. Request voice call
8. Talk to AI assistant
9. End conversation
10. Receive follow up information

The Lead cannot:

1. View internal score
2. View RM portal
3. View admin dashboard
4. Access other lead data

### 3.2 Admin

---

The Admin is the client side manager.

The Admin can:

1. Log in
2. View dashboard
3. View all leads
4. View lead details
5. View conversations
6. View score and classification
7. View transcripts
8. View objections
9. View RM assignment

10. View follow ups
11. View analytics
12. Manage users and RMs
13. View campaigns
14. View knowledge documents
15. View feedback
16. Deactivate RMs
17. Add RMs

The Admin cannot:

1. Directly edit AI model internals in MVP
2. Directly access database tables from frontend
3. Send real WhatsApp API messages in MVP unless integration is added

## 3.3 RM

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The RM is the relationship manager or sales person.

The RM can:

1. Log in
2. View assigned Hot leads
3. View assigned leads
4. Open lead handoff details
5. Read transcript
6. Read AI summary
7. View objections
8. View score reason
9. View suggested opening line
10. Mark task as in progress
11. Mark lead converted
12. Mark lead closed
13. Open WhatsApp follow up link
14. Mark follow up sent
15. Submit feedback on AI scoring

The RM cannot:

1. View all client leads unless assigned
2. Manage users
3. Manage campaigns
4. View unrelated RM tasks
5. Access admin only analytics if restricted

## 3.4 AI Agent

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The AI Agent is the automated conversation actor.

The AI Agent can:

1. Start chat or voice conversation
2. Use selected language
3. Switch language when requested
4. Explain partner program
5. Handle objections
6. Ask qualification questions
7. Save conversation transcript
8. Save language preference
9. Save callback timing
10. Save WhatsApp consent
11. End conversation

The AI Agent cannot:

1. Guarantee income
2. Give investment advice
3. Ask for passwords or OTPs
4. Invent unsupported claims
5. Make final business decision without scoring pipeline
6. Replace RM completely

## 3.5 System

---

The System handles backend automation.

The System can:

1. Create lead records
2. Create conversations
3. Store messages
4. Compile transcripts
5. Run scoring
6. Classify leads
7. Create RM tasks

8. Generate WhatsApp links
  9. Create notifications
  10. Update analytics
- 

## 4. Product Scope

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### 4.1 In Scope for MVP

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The MVP includes:

1. LeadOS SaaS marketing website
2. Dummy client website
3. Lead form
4. Public lead creation
5. Chat based AI conversation
6. Voice request or Vapi voice flow if ready
7. Language selection
8. Basic language switching
9. AI objection handling
10. Transcript storage
11. Lead scoring
12. Hot, Warm, Cold classification
13. RM handoff for Hot leads
14. WhatsApp link generation for Warm leads
15. Cold lead closure
16. Admin portal
17. RM portal
18. Users page
19. Analytics pages
20. Follow ups page
21. Knowledge base page
22. Feedback page
23. Supabase Auth
24. Supabase Postgres
25. Supabase Edge Functions
26. OpenAI based conversation and scoring
27. Mock or real Vapi voice path

### 4.2 Out of Scope for MVP

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The MVP does not include:

1. Full CRM replacement
  2. Real WhatsApp Business API approval
  3. Payment or subscription billing
  4. Multi client production onboarding flow
  5. Complex role hierarchy
  6. Advanced AI model training
  7. Automatic prompt deployment from RM feedback
  8. Real phone number purchase and telephony if not ready
  9. Complex campaign builder
  10. Advanced audit trails
  11. Mobile app
  12. Full data chatbot for all portals unless time allows
  13. Production grade n8n workflows if backend can handle MVP routing
- 

## 5. System Architecture Overview

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### 5.1 High Level Architecture

---

```
LeadOS SaaS Website
Dummy Client Website
Admin Portal
RM Portal
    ↓
Supabase Edge Function APIs
    ↓
Service Layer
    ↓
Supabase Postgres
    ↓
AI Layer and Automation Layer
```

## 5.2 Main Backend Components

---

1. Auth Service
2. Lead Service
3. Conversation Service
4. Chat Agent Service
5. Voice Service
6. Knowledge Service
7. Memory Service
8. Scoring Service
9. Routing Service
10. RM Task Service
11. Follow Up Service
12. Notification Service
13. Analytics Service
14. User Service
15. Feedback Service

## 5.3 External Integrations

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1. Supabase Auth
2. Supabase Postgres
3. Supabase Edge Functions
4. OpenAI
5. Vapi
6. WhatsApp deep link through wa.me
7. n8n optional for automation
8. Mem0 optional for memory

# 6. Website Requirements

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## 6.1 LeadOS SaaS Website

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### 6.1.1 Purpose

The SaaS website should present LeadOS as a real enterprise SaaS product.

### 6.1.2 Required Pages or Sections

1. Hero section
2. Problem section
3. Solution section
4. How it works section
5. Features section
6. Product preview section
7. Architecture preview section
8. Demo CTA section
9. Footer

### 6.1.3 Required CTAs

The website must provide buttons for:

1. Open Dummy Client Demo
2. Open Admin Portal
3. Open RM Portal

## 6.1.4 Content Requirements

The SaaS website must explain:

1. Lead capture
2. AI conversation
3. Multilingual handling
4. Objection handling
5. Lead scoring
6. RM handoff
7. WhatsApp follow up
8. Dashboard analytics

## 6.1.5 Branding Requirement

The website must not use Rupeezy branding.

It should use the LeadOS brand and a fictional demo client.

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## 6.2 Dummy Client Website

### 6.2.1 Purpose

The dummy client website simulates the lead source.

The fictional client name should be:

FinPartner Pro

### 6.2.2 Required Sections

1. Hero section
2. Benefits section
3. Lead form
4. Chat or voice start section
5. Thank you state

### 6.2.3 Lead Form Fields

The form must collect:

1. Name
2. Phone
3. Email optional
4. City optional
5. Role
6. Network size optional
7. Preferred language
8. Preferred channel
9. WhatsApp consent

### 6.2.4 Role Options

1. Financial Advisor
2. MFD
3. Insurance Agent
4. Finance Influencer
5. Referrer
6. Other

### 6.2.5 Language Options

1. English
2. Hindi
3. Hinglish

### 6.2.6 Channel Options

1. Chat
2. Voice

### 6.2.7 Form Behavior

When the lead submits the form:

```
Validate form
→ Create lead
→ Store lead session token
→ Start chat if channel is chat
→ Show voice start state if channel is voice
```

## 6.2.8 Completion Behavior

After conversation ends, show:

```
Thanks for speaking with our AI assistant. If you are a good fit, a relationship manager will follow up with the next steps.
```

The lead should not see internal Hot, Warm, Cold classification.

# 7. Admin Portal Requirements

## 7.1 Purpose

The Admin Portal is the control center for the client admin.

It must show:

1. Lead funnel
2. AI conversation outcomes
3. Hot, Warm, Cold breakdown
4. RM assignments
5. Follow ups
6. Analytics
7. Users
8. Feedback
9. Knowledge documents

## 7.2 Admin Routes

Required routes:

```
/admin/login
/admin/dashboard
/admin/leads
/admin/leads/:leadId
/admin/conversations
/admin/analytics
/admin/campaigns
/admin/users
/admin/followups
/admin/knowledge
/admin/feedback
/admin/settings
```

## 7.3 Admin Login

The admin must be able to log in using Supabase Auth.

Demo credentials can be seeded.

## 7.4 Admin Dashboard

Dashboard must show KPI cards:

1. Total Leads
2. Conversations Completed
3. Hot Leads
4. Warm Leads
5. Cold Leads
6. Assigned to RM
7. Follow Ups Ready
8. Converted

Dashboard must show charts:

1. Lead funnel
2. Hot, Warm, Cold distribution
3. Language distribution
4. Objection distribution

Dashboard must show lists:

1. Recent conversations
2. Recent Hot leads
3. Follow up summary
4. RM performance preview

## 7.5 Leads List

---

The Leads page must show:

1. Lead name
2. Phone
3. City
4. Role
5. Language
6. Channel
7. Status
8. Classification
9. Score
10. Assigned RM
11. Created date
12. Actions

Filters:

1. Search
2. Status
3. Classification
4. Language
5. Campaign
6. Assigned RM
7. Source
8. Date range

Sorting:

1. Created date
2. Updated date
3. Score
4. Name
5. Status
6. Classification

## 7.6 Lead Detail

---

Lead detail must show:

1. Lead profile
2. Current status
3. Classification
4. Latest score
5. Preferred language
6. Preferred channel
7. WhatsApp consent
8. Campaign
9. Assigned RM
10. AI summary
11. Score reason
12. Readiness score
13. Engagement score
14. Fit score
15. Positive signals
16. Negative signals
17. Objections
18. Objection resolution status
19. Transcript
20. RM task
21. Suggested RM opening line
22. Follow up card for Warm leads

## 7.7 Conversations Page

---

The Conversations page must show:

1. Lead name
2. Channel



3. Language
4. Status
5. Duration
6. Classification
7. Score
8. Started time
9. Action

Filters:

1. Channel
2. Language
3. Status
4. Classification
5. Date range

## 7.8 Analytics Page

---

Analytics must show:

1. Lead funnel
2. Classification distribution
3. Language distribution
4. Objection distribution
5. RM performance
6. Campaign performance
7. Follow up performance

## 7.9 Campaigns Page

---

Campaigns must show:

1. Campaign name
2. Source
3. Total leads
4. Hot leads
5. Warm leads
6. Cold leads
7. Converted leads
8. Active status

## 7.10 Users Page

---

Users page must show all admins and RMs.

It must show:

1. Name
2. Email
3. Role
4. Active status
5. Assigned leads
6. Pending tasks
7. Converted leads
8. Created date
9. Actions

Admin must be able to:

1. Add RM
2. Deactivate RM
3. View user detail
4. Filter by role
5. Filter by active status
6. Search by name or email

For MVP, deactivation is preferred over permanent delete.

## 7.11 Follow Ups Page

---

Follow Ups must show Warm lead follow ups.

Fields:

1. Lead name
2. Phone
3. Classification
4. Channel
5. Status
6. Message preview
7. WhatsApp link
8. Created time
9. Actions

Actions:

1. Open WhatsApp
2. Mark Sent
3. Cancel optional

## 7.12 Knowledge Base Page

---

Knowledge page must show:

1. Sales script
2. FAQ
3. Onboarding guide
4. Objection guide

For MVP, editing is optional. Viewing is enough.

## 7.13 Feedback Page

---

Feedback page must show RM feedback on AI scoring.

Fields:

1. Lead name
2. RM name
3. Original classification
4. Corrected classification
5. Original score
6. Corrected score
7. Feedback type
8. Comment
9. Created date

## 7.14 Settings Page

---

Settings page must show:

1. Client info
2. Supported languages
3. Scoring thresholds
4. Voice provider
5. LLM provider
6. RAG status
7. Memory status
8. Automation status

---

# 8. RM Portal Requirements

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## 8.1 Purpose

---

The RM Portal helps RMs work on assigned leads.

It must prove that Hot leads are handed off with full context.

## 8.2 RM Routes

---

Required routes:

```
/rm/login
/rm/dashboard
/rm/hot-leads
/rm/leads
/rm/leads/:leadId
/rm/followups
/rm/notifications
/rm/feedback
```

Optional route:

```
/rm/performance
```

## 8.3 RM Dashboard

---

RM Dashboard must show:

1. Assigned Hot Leads
2. Pending Tasks
3. Follow Ups Due
4. Converted Leads
5. Closed Leads
6. Hot lead queue preview
7. Follow ups due
8. Recent handoffs
9. Notifications preview

## 8.4 Hot Leads Page

---

Hot Leads page must show only Hot leads assigned to the RM.

Each lead should show:

1. Lead name
2. Phone
3. City
4. Role
5. Language
6. Score
7. Main objection
8. Recommended next action
9. Task status
10. Assigned time
11. Open lead action

## 8.5 My Leads Page

---

My Leads page must show all assigned leads.

Columns:

1. Lead name
2. Phone
3. Language
4. Status
5. Classification
6. Score
7. Recommended action
8. Created date
9. Actions

## 8.6 RM Lead Detail

---

RM Lead Detail must show:

1. Lead profile
2. AI score
3. Score reason
4. Readiness score
5. Engagement score
6. Fit score
7. Positive signals
8. Negative signals
9. AI summary
10. Objections
11. Transcript
12. Recommended next action
13. Suggested RM opening line
14. Task actions
15. Follow up section
16. Feedback section

RM task actions:

1. Mark In Progress
2. Mark Converted
3. Mark Closed
4. Reschedule Follow Up optional
5. Add Note optional

## 8.7 RM Follow Ups Page

---

RM Follow Ups must show follow ups linked to assigned leads.

Actions:

1. Open WhatsApp
2. Mark Sent
3. Reschedule optional
4. Cancel optional

## 8.8 Notifications Page

---

Notifications must show:

1. New Hot Lead Assigned
2. Follow Up Due
3. Score Ready
4. Task Updated

Actions:

1. Mark read
2. Mark all read
3. Open related lead

## 8.9 Feedback Page

---

RM must be able to submit feedback on AI scoring.

Fields:

1. Lead
2. Corrected classification
3. Corrected score optional
4. Feedback type
5. Comment

Feedback types:

1. Score Wrong
  2. Objection Wrong
  3. Bad Response
  4. Good Response
  5. Other
- 

# 9. AI Conversation Requirements

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## 9.1 Purpose

---

The AI Agent must conduct structured sales conversations.

It must:

1. Start naturally
2. Use lead's language
3. Explain partner program
4. Handle objections
5. Ask qualification questions
6. Close with clear next step
7. Produce transcript for scoring

## 9.2 Supported Channels

---

1. Chat
2. Voice through Vapi if ready

## 9.3 Supported Languages

---

1. English
2. Hindi
3. Hinglish

## 9.4 Language Detection and Switching

---

The system must choose language using:

1. Selected language from form
2. Previous lead memory
3. First user message

4. Voice transcript cues
5. Clarifying question

The agent must switch language if lead asks.

Example:

```
Lead: Hindi mein samjha do
AI: Haan bilkul, main Hinglish mein explain karta hoon.
```

## 9.5 Conversation Stages

---

The agent must follow these stages:

1. Greeting
2. Language Confirmation
3. Hook
4. Benefit Pitch
5. Discovery
6. Objection Handling
7. Qualification
8. Close
9. End

## 9.6 Required Benefits to Explain

---

The AI must explain these benefits when present in approved knowledge:

1. Zero joining fee
2. Strong revenue share
3. Fast or daily payouts
4. Partner support
5. Easy onboarding

## 9.7 Required Objections

---

The AI must handle five core objections:

1. Already with another broker
2. Not enough contacts
3. Client support concern
4. Trust concern
5. Think later or call later

## 9.8 Objection Handling Pattern

---

For every objection, the agent must:

1. Acknowledge concern
2. Understand actual reason
3. Answer using approved knowledge
4. Personalize using previous context
5. Ask next step question

## 9.9 AI Restrictions

---

The AI must not:

1. Guarantee income
2. Give investment advice
3. Ask for OTP or password
4. Invent regulatory claims
5. Insult competitors
6. Pressure the lead
7. Ignore do not contact intent

---

# 10. Scoring Requirements

---

## 10.1 Scoring Trigger

---

Scoring must run automatically after conversation completion.

For chat:

```
Chat ended
→ transcript compiled
→ scoring runs
```

For voice:

```
Vapi call ended webhook
→ transcript saved
→ scoring runs
```

## 10.2 Score Dimensions

---

Total score must be out of 100.

Score breakdown:

1. Readiness, 40 points
2. Interest and Engagement, 35 points
3. Fit and Network, 25 points

## 10.3 Classification Thresholds

---

Hot:

```
Score 70 or more with clear next step intent
```

Warm:

```
Score between 40 and 69 or interested but not ready
```

Cold:

```
Score below 40 or clearly not interested
```

## 10.4 Override Rules

---

Force Hot when:

1. Lead asks for RM callback soon
2. Lead asks how to onboard
3. Lead wants to start
4. Lead gives clear next step intent

Force Warm when:

1. Lead asks for details first
2. Lead wants to verify
3. Lead gives callback timing but not strong intent
4. Lead accepts WhatsApp follow up

Force Cold when:

1. Lead says not interested
2. Lead asks not to be contacted
3. Lead refuses follow up
4. Lead is invalid or unrelated

## 10.5 Scoring Output

---

Scoring must produce:

1. Classification
2. Total score
3. Readiness score
4. Engagement score
5. Fit score
6. Reason
7. Recommended next action
8. Suggested RM opening line
9. Positive signals
10. Negative signals
11. Objections
12. Extracted data

## 10.6 Objection Extraction

---

For every objection, extract:

1. Type
2. Lead statement
3. Resolution status
4. AI response summary
5. Impact on score

Resolution statuses:

1. Resolved
  2. Partially Resolved
  3. Unresolved
- 

## 11. Routing Requirements

---

### 11.1 Hot Lead Routing

---

When a lead is Hot:

```
Update lead classification to hot
→ Update status to assigned_to_rm
→ Assign RM
→ Create RM task
→ Create notification
```

### 11.2 Warm Lead Routing

---

When a lead is Warm:

```
Update lead classification to warm
→ Update status to follow_up_scheduled
→ Generate WhatsApp follow up message
→ Generate wa.me link
→ Create follow up record
```

### 11.3 Cold Lead Routing

---

When a lead is Cold:

```
Update lead classification to cold
→ Update status to closed
→ No RM task
→ No follow up by default
```

---

## 12. Data Requirements

---

### 12.1 Main Tables

---

The system must use these tables:

1. clients
2. users
3. campaigns
4. leads
5. conversations
6. conversation\_messages
7. lead\_scores
8. rm\_tasks
9. follow\_ups
10. notifications

11. knowledge\_documents
12. rm\_feedback

Optional:

1. knowledge\_chunks

## 12.2 Lead Status Values

---

Allowed lead statuses:

```
new
pending_contact
conversation_completed
assigned_to_rm
follow_up_scheduled
converted
closed
```

## 12.3 Lead Classification Values

---

Allowed classifications:

```
hot
warm
cold
```

## 12.4 User Roles

---

Allowed roles:

```
admin
rm
```

## 12.5 Conversation Channels

---

Allowed channels:

```
chat
voice
```

## 12.6 Follow Up Status

---

Allowed follow up statuses:

```
ready
opened
sent
cancelled
```

## 12.7 RM Task Status

---

Allowed RM task statuses:

```
pending
in_progress
converted
closed
```

---

## 13. API Requirements



---

## 13.1 API Architecture

---

Frontend must not call Supabase DB directly.

All frontend apps must call Supabase Edge Function APIs.

```
Frontend
→ Edge Function API
→ Auth and role checks
→ Service layer
→ Supabase Postgres
```

## 13.2 Auth APIs

---

Required:

```
POST /auth/login
GET /auth/me
POST /auth/logout
POST /auth/refresh
```

## 13.3 Public APIs

---

Required:

```
GET /public/client/:clientSlug
POST /public/leads
```

## 13.4 Lead APIs

---

Required:

```
GET /leads
GET /leads/:leadId
PATCH /leads/:leadId
POST /leads/:leadId/assign-rm
```

## 13.5 Chat APIs

---

Required:

```
POST /chat/start
POST /chat/message
POST /chat/end
```

## 13.6 Voice APIs

---

Required:

```
POST /voice/start
POST /vapi-webhook
```

## 13.7 Scoring APIs

---

Required:

```
POST /scores/run
GET /leads/:leadId/scores
```

## 13.8 RM Task APIs

---

Required:

```
GET /rm/tasks
GET /rm/tasks/:taskId
PATCH /rm/tasks/:taskId
```

## 13.9 Follow Up APIs

---

Required:

```
GET /followups
POST /followups/:followUpId/open
POST /followups/:followUpId/mark-sent
```

## 13.10 Notification APIs

---

Required:

```
GET /notifications
PATCH /notifications/:notificationId/read
PATCH /notifications/read-all
```

## 13.11 Analytics APIs

---

Required:

```
GET /analytics/overview
GET /analytics/funnel
GET /analytics/languages
GET /analytics/objections
GET /analytics/rm-performance
```

## 13.12 User APIs

---

Required:

```
GET /users
GET /users/rms
POST /users/rms
PATCH /users/:userId
```

## 13.13 Knowledge APIs

---

Required:

```
GET /knowledge
GET /knowledge/:documentId
POST /knowledge
```

POST is optional for MVP if docs are seeded manually.

## 13.14 Feedback APIs

---

Required:

```
GET /feedback
POST /feedback
```

---

## 14. Response Format Requirements

---

### 14.1 Success Response

---

```
{
  "success": true,
  "data": {},
  "meta": {}
}
```

### 14.2 List Response

---

```
{
  "success": true,
  "data": [],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 100,
    "totalPages": 5
  }
}
```

### 14.3 Error Response

---

```
{
  "success": false,
  "error": {
    "code": "LEAD_NOT_FOUND",
    "message": "Lead not found",
    "details": {}
  },
  "requestId": "req_123"
}
```

---

## 15. Security Requirements

---

### 15.1 Authentication

---

Admin and RM users must authenticate through Supabase Auth.

### 15.2 Authorization

---

Role based access must be enforced in backend APIs.

Admin can access all client data.

RM can access only assigned lead data.

Public lead can access only their own chat session through lead session token.

### 15.3 Data Access

---

Frontend must not call Supabase tables directly.

All reads and writes must go through Edge Functions.

### 15.4 RLS

---

---

RLS must be enabled on all tables.

Direct anonymous table access must be blocked.

## 15.5 Secrets

---

The following must never be exposed to frontend:

1. Supabase service role key
2. OpenAI API key
3. Vapi API key
4. Webhook secrets
5. n8n private webhook secrets

## 15.6 Webhook Security

---

Vapi webhook must be protected using a webhook secret or signature verification.

## 15.7 Privacy

---

The system must avoid collecting unnecessary sensitive data.

The AI must not ask for:

1. OTP
2. Password
3. Banking credentials
4. Sensitive investment details

---

# 16. Non Functional Requirements

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## 16.1 Performance

---

The system should:

1. Load dashboard within acceptable demo time
2. Paginate lead lists
3. Avoid loading all transcripts unnecessarily
4. Show loading states during AI responses
5. Avoid freezing on long transcripts

## 16.2 Reliability

---

The system should:

1. Save transcript before scoring
2. Keep lead data even if scoring fails
3. Use fallback scoring if AI fails
4. Not crash if Vapi fails
5. Allow chat fallback if voice fails

## 16.3 Usability

---

The UI should be:

1. Clean
2. Modern
3. Easy to scan
4. Demo friendly
5. Clear for non technical judges

## 16.4 Maintainability

---

Code should use:

1. Reusable components
2. Typed models
3. Service functions
4. Clean API contracts
5. Centralized constants
6. Clear folder structure

## 16.5 Scalability

---

The MVP should be designed so later it can support:

1. Multiple clients
2. More languages
3. More campaigns
4. More RMs
5. Real WhatsApp API
6. Pipecat scale path
7. Better RAG
8. Memory improvements

---

## 17. Integration Requirements

### 17.1 Supabase

---

Supabase will provide:

1. Auth
2. Postgres
3. Edge Functions

### 17.2 OpenAI

---

OpenAI will provide:

1. AI conversation generation
2. Scoring JSON output
3. Structured extraction

### 17.3 Vapi

---

Vapi will provide:

1. Voice session
2. Voice transcript events
3. Call ended webhook

For MVP, Vapi can be demonstrated or replaced with chat fallback if unstable.

### 17.4 WhatsApp

---

MVP uses wa.me links.

No Meta Cloud API approval is required for MVP.

### 17.5 n8n

---

n8n can be used for automation if ready.

If not ready, backend routing service must handle:

1. RM task creation
2. Follow up creation
3. Notification creation

---

## 18. Functional Requirement List

### 18.1 Lead Capture

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FR1. System shall allow a lead to submit a form from dummy client website. FR2. System shall validate required lead form fields. FR3. System shall create a lead record after successful submission. FR4. System shall assign client and campaign to the lead. FR5. System shall generate public lead session token. FR6. System shall store preferred language. FR7. System shall store preferred channel. FR8. System shall store WhatsApp consent.

### 18.2 AI Chat

---

FR9. System shall start chat conversation for a valid lead session. FR10. System shall create conversation record. FR11. System shall save assistant greeting. FR12.

System shall save user messages. FR13. System shall generate assistant responses. FR14. System shall save assistant responses. FR15. System shall support selected language. FR16. System shall support language switching. FR17. System shall end conversation. FR18. System shall compile transcript.

## 18.3 AI Voice

---

FR19. System shall create voice conversation when requested. FR20. System shall receive Vapi transcript events. FR21. System shall save voice transcript messages. FR22. System shall receive call ended event. FR23. System shall complete conversation after call ends. FR24. System shall trigger scoring after voice call ends.

## 18.4 Scoring

---

FR25. System shall run scoring after conversation completion. FR26. System shall score readiness out of 40. FR27. System shall score engagement out of 35. FR28. System shall score fit out of 25. FR29. System shall calculate total score out of 100. FR30. System shall classify lead as Hot, Warm, or Cold. FR31. System shall extract objections. FR32. System shall generate score reason. FR33. System shall generate recommended next action. FR34. System shall generate suggested RM opening line for Hot leads. FR35. System shall save score in database.

## 18.5 Routing

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FR36. System shall create RM task for Hot leads. FR37. System shall assign RM to Hot lead. FR38. System shall create notification for assigned RM. FR39. System shall generate WhatsApp follow up for Warm leads. FR40. System shall generate wa.me link for Warm leads. FR41. System shall close Cold leads.

## 18.6 Admin Portal

---

FR42. Admin shall log in. FR43. Admin shall view dashboard. FR44. Admin shall view all leads. FR45. Admin shall filter leads. FR46. Admin shall sort leads. FR47. Admin shall view lead detail. FR48. Admin shall view transcript. FR49. Admin shall view score and reason. FR50. Admin shall view objections. FR51. Admin shall view follow ups. FR52. Admin shall view analytics. FR53. Admin shall view users. FR54. Admin shall add RM. FR55. Admin shall deactivate RM. FR56. Admin shall view knowledge documents. FR57. Admin shall view feedback.

## 18.7 RM Portal

---

FR58. RM shall log in. FR59. RM shall view assigned Hot leads. FR60. RM shall view assigned lead details. FR61. RM shall view transcript. FR62. RM shall view AI summary. FR63. RM shall view score reason. FR64. RM shall view suggested opening line. FR65. RM shall mark lead converted. FR66. RM shall mark lead closed. FR67. RM shall open WhatsApp follow up link. FR68. RM shall mark follow up sent. FR69. RM shall submit feedback.

## 18.8 Analytics

---

FR70. System shall show total leads. FR71. System shall show conversations completed. FR72. System shall show Hot lead count. FR73. System shall show Warm lead count. FR74. System shall show Cold lead count. FR75. System shall show language distribution. FR76. System shall show objection distribution. FR77. System shall show RM performance. FR78. System shall show campaign performance.

---

# 19. MVP Acceptance Criteria

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The MVP is accepted if:

1. SaaS website opens and explains product
2. Dummy client website captures lead
3. Lead is stored in backend
4. Chat starts for lead
5. AI responds naturally
6. AI handles at least one required objection
7. Conversation ends
8. Transcript is saved
9. Scoring runs
10. Lead is classified Hot, Warm, or Cold
11. Hot lead creates RM task
12. Warm lead creates WhatsApp link
13. Cold lead is closed
14. Admin dashboard shows metrics
15. Admin lead detail shows score, transcript, objections, and next action
16. RM portal shows Hot lead with full handoff context
17. RM can mark lead converted or closed
18. Follow up page opens WhatsApp link
19. Role based access works
20. Demo path works without manual database editing

---

# 20. Main Demo Scenario

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## 20.1 Hot Lead Demo

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Lead data:

```
Name: Amit Sharma
Language: Hinglish
Role: Financial Advisor
Network Size: 25 to 30
Channel: Chat
WhatsApp Consent: Yes
```

Conversation:

```
Lead says: I already have another broker.
AI handles existing broker objection.
Lead says: Send WhatsApp and RM can call today.
Conversation ends.
```

Expected result:

```
Classification: Hot
Lead status: assigned_to_rm
RM task created
Notification created
RM portal shows lead
Lead detail shows transcript, score reason, objection, suggested opening line
```

## 20.2 Warm Lead Demo

---

Lead says:

```
I need to think. Send details.
```

Expected result:

```
Classification: Warm
Lead status: follow_up_scheduled
WhatsApp link created
Follow up visible in Admin and RM portal
```

## 20.3 Cold Lead Demo

---

Lead says:

```
Not interested. Do not contact me.
```

Expected result:

```
Classification: Cold
Lead status: closed
No RM task
No follow up
```

---

# 21. Implementation Priority

## 21.1 Priority 1

---

Must finish first:

1. Backend database
2. Auth
3. Public lead creation
4. Chat start

5. Chat message
6. Chat end
7. Transcript storage
8. Scoring
9. Routing
10. Admin lead list
11. Admin lead detail
12. RM Hot lead detail

## 21.2 Priority 2

---

Finish next:

1. Dashboard analytics
2. Follow ups
3. Users page
4. Notifications
5. Feedback
6. Voice request path
7. Vapi webhook if ready

## 21.3 Priority 3

---

Only after core works:

1. Knowledge editing
2. Campaign creation
3. Data chatbot
4. Advanced charts
5. Real n8n workflows
6. Extra animations
7. Pipecat scale path demo

---

# 22. Risk Handling

---

## 22.1 AI Hallucination

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Mitigation:

1. Use strict prompt
2. Use approved knowledge documents
3. Instruct AI not to invent claims
4. Fallback to RM confirmation when unsure

## 22.2 Wrong Scoring

---

Mitigation:

1. Explain score
2. Store score breakdown
3. Allow RM feedback
4. Use validation rules
5. Use fallback scoring

## 22.3 Language Detection Error

---

Mitigation:

1. Use selected language first
2. Allow user to switch
3. Ask clarifying question if uncertain

## 22.4 Voice Failure

---

Mitigation:

1. Chat is primary stable demo path
2. Voice has fallback to chat
3. Vapi webhook can be demonstrated if ready

## 22.5 WhatsApp API Delay

---

Mitigation:



1. Use wa.me link for MVP
2. Show message preview
3. Allow manual open and mark sent

## 22.6 Time Constraint

---

Mitigation:

1. Freeze MVP scope
  2. Prioritize core flow
  3. Use seeded data for backup
  4. Avoid overbuilding
- 

## 23. Final Product Success Definition

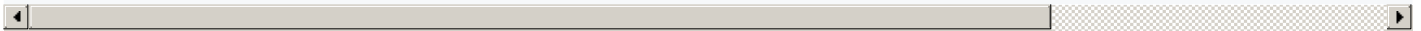
---

LeadOS is successful if a judge can clearly understand this within the demo:

This is not just a chatbot.

It is a complete AI lead conversion system.

It captures leads, talks to them in their language, handles objections, scores intent, routes Hot leads to RMs, sends Warm follow up



The system must show that RMs receive useful context, not just contact details.

That is the core winning point.